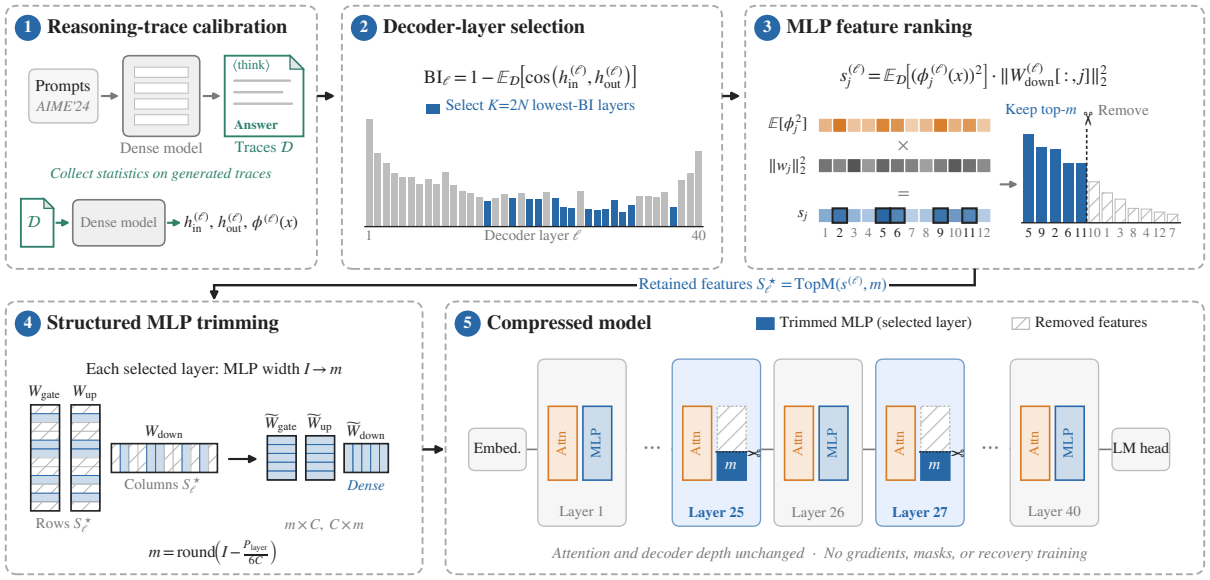
1 Reasoning-trace calibration
Prompts
AIME'24
Dense model
⟨think⟩
Answer
Traces D
Collect statistics on generated traces
D
Dense model
$h_{\text{in}}^{(\ell)}, h_{\text{out}}^{(\ell)}, \phi^{(\ell)}(x)$
2 Decoder-layer selection
$\text{BI}_\ell = 1 - \mathbb{E}_D\left[\cos\left(h_{\text{in}}^{(\ell)}, h_{\text{out}}^{(\ell)}\right)\right]$
Select $K{=}2N$ lowest-BI layers
1
Decoder layer ℓ
40
3 MLP feature ranking
$s_j^{(\ell)} = \mathbb{E}_D\left[(\phi_j^{(\ell)}(x))^2\right] \cdot \|W_{\text{down}}^{(\ell)}[:, j]\|_2^2$
$\mathbb{E}[\phi_j^2]$
$\|w_j\|_2^2$
s_j
1 2 3 4 5 6 7 8 9 10 11 12
Keep top-m
Remove
5 9 2 6 11 10 1 3 8 4 12 7
Retained features $S_\ell^\star = \text{TopM}(s^{(\ell)}, m)$
4 Structured MLP trimming
Each selected layer: MLP width $I \to m$
W_{gate}
W_{up}
W_{down}
Columns $S_\ell^\star$
Rows $S_\ell^\star$
$\widetilde{W}_{\text{gate}}$
$\widetilde{W}_{\text{up}}$
$\widetilde{W}_{\text{down}}$
Dense
$m \times C,\ C \times m$
$m = \text{round}\left(I - \frac{P_{\text{layer}}}{6C}\right)$
5 Compressed model
Trimmed MLP (selected layer)
Removed features
Embed.
Attn
MLP
Layer 1
Layer 25
Layer 26
Layer 27
Layer 40
LM head
Attention and decoder depth unchanged · No gradients, masks, or recovery training